

DepthWizard demo video: stitch script

SIH26175 · Team Dev Up (129655) · about 2:50 · every clip is in `D:\sih2026\video\`

1. What you still need to record

Only the voice-over (section 3). Every clip is already recorded. Record each segment as its own take, in a quiet room, phone about 20 cm from your mouth; separate takes make it easy to line each one up under its clip. Measuring and offline use are not shown as shots; both are mentioned in the voice-over (segments 4 and 9).

2. The sequence

#	Clip	Length	In final video	Shows
0	HAVE c0_intro_card.mp4	9 s	0:00–0:09	Title card: DepthWizard, SIH26175, Team Dev Up
1	HAVE c1_hook_sikkim.mp4	15 s	0:09–0:24	Orbit over the Sikkim valley, no panels
2	HAVE c2_problem_omaha.mp4	20 s	0:24–0:44	Omaha flat from above, tilting into 3D
3	HAVE c3_upload.mp4	47 s	0:44–1:31	Your upload, wait already cut out
4	HAVE c4_flythrough_jacksonville.mp4	15 s	1:31–1:46	Low fly-through over the park
5	HAVE c5a_confidence_jacksonville.mp4	12 s	1:46–1:58	Switches to "How sure we are" at ~4 s
6	HAVE c5b_lidar_compare_omaha.mp4	14 s	1:58–2:12	Divider sweeps: our model vs real LiDAR
7	HAVE c5c_where_wrong_omaha.mp4	10 s	2:12–2:22	"Where we're wrong" colours
8	HAVE c6_flood_sikkim.mp4	18 s	2:22–2:40	Water rises to 1,143 m, readout climbs
9	HAVE c8_end_card.mp4	8 s	2:40–2:48	Title, team, links, Maxar credit

Editing rules. Cut straight from one clip to the next; a 0.3 s cross-fade is fine everywhere. Lay each voice take so it starts about half a second into its clip. If a take is shorter than the clip, trim the clip's end; if longer, slow the clip slightly or hold its last frame. Mute every clip's own audio. Put "Sikkim imagery © Maxar Technologies, CC-BY-4.0" in the video description.

3. Clip by clip: what plays, and what you say

0 · Intro — c0_intro_card.mp4

0:00–0:09

Title card, fading in and out: Smart India Hackathon 2026 · ISRO · SIH26175, DepthWizard, the problem statement's title, Team Dev Up, team ID 129655, theme Disaster Management.

Hello, we're Team Dev Up. This is DepthWizard — our solution to ISRO's problem statement on single-view height estimation and 3D flythrough.

1 · Hook — c1_hook_sikkim.mp4

0:09–0:24

Slow orbit and push-in over the Sikkim valley. No interface on screen.

One satellite image. No stereo pair, no laser survey, no new flight. This is Sikkim — rebuilt in 3D from a single picture, by DepthWizard.

2 · The problem — c2_problem_omaha.mp4

0:24–0:44

Opens straight down on Omaha (it looks like a flat photo), then tilts into 3D from about 4 s.

From above, a satellite image is flat. Height normally needs stereo imagery or LiDAR — expensive, and impossible for the decades of archive India already holds. DepthWizard estimates height in metres from one image, and tells you how sure it is.

3 · Upload — c3_upload.mp4

0:44–1:31

0–7 s the sample window opens · 7–11 s a file is chosen from the computer · 11–20 s the progress bar (the minute of processing is already cut) · 20–47 s the result opens and is explored, download links in the left panel.

0–11 s, while choosing

Anyone can try it. Pick one of our sample images, or any image from your own computer — a GeoTIFF, a PNG or a JPG.

11–20 s, over the progress bar

It runs on an ordinary CPU server, with no GPU. This image takes about a minute; we've cut the wait.

20–47 s, over the result

Because this file carries coordinates, the heights are anchored to open Copernicus terrain — a real surface in metres above sea level, exactly what ISRO evaluates. Every result downloads as standard GeoTIFFs: the DSM, height above ground, and a per-pixel uncertainty map, ready for QGIS — plus a quick-look picture anyone can open.

4 · Explore and measure — c4_flythrough_jacksonville.mp4

1:31–1:46

Low, sweeping fly-through over a park in Jacksonville, interface visible.

Fly through any scene freely. Click two points to measure the distance on the ground, the height difference, and how sure the model is. It opens at true scale — nothing is stretched.

5 · Confidence — c5a_confidence_jacksonville.mp4

1:46–1:58

Satellite view, switching to the "How sure we are" colours at about 4 s.

Every height comes with a confidence. Warm colours are where the model is unsure — and that confidence is checked against real LiDAR.

6 · Checked against LiDAR — c5b_lidar_compare_omaha.mp4

1:58–2:12

The divider sweeps left and right: our model on one side, the real laser survey on the other.

Drag to compare with real airborne LiDAR. On regions held out from training, we're within 3.5 metres per building; the published figure over Asia is 5.9.

7 · Where we're wrong — c5c_where_wrong_omaha.mp4

2:12–2:22

Error colours on Omaha: blue is too low, red too high.

And we show where we're wrong. Blue is too low — very tall towers are our known weakness, and the viewer says so.

8 · Disaster use case — c6_flood_sikkim.mp4

2:22–2:40

"Flood the valley" is on; the water rises for about 14 s to 1,143 m, then holds on the readout.

Built for disaster management. Raise the water in a Sikkim valley: at 1,143 metres, a fifth of the valley and 44 buildings are under water — and buildings too close to call are flagged.

8 s; hold the last frame a few seconds if this take runs long. Title card: DepthWizard, Team Dev Up, SIH26175, team ID 129655, live / docs / code links, Maxar credit.

No internet? The whole viewer also ships as one HTML file that runs offline.
DepthWizard — height from one satellite image, with the evidence to trust it.

Every number spoken is on screen in the viewer or on deck slide 7: calibration and 3.5 m per building (held-out DFC2019), 5.9 m published peer, 1,143 m / 20.5 % / 44 buildings (live flood tool). Do not add others.